An energy budget

antocciphysics | TEACHER KEY | CPA + MATCHING SEI

Learning target: Connect work, motion, heating and an honest model conclusion.

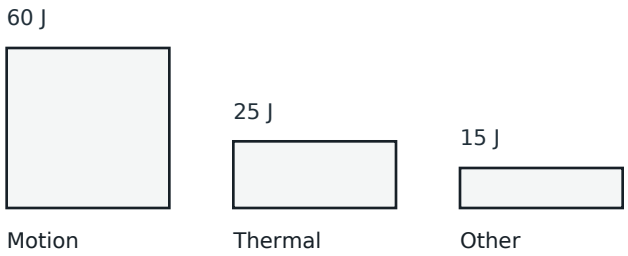
Accept equivalent correct methods and clearly labeled models. Language supports do not change the physics demand. Supplied data are examples; actual lab measurements and teacher checkpoints must be recorded from the lesson.

01. Warm-up: bring back an earlier idea *Review: Unit 4, Day 18.*
3 J remains unaccounted. Seek another storage/transfer or check measurements.

02. Worked example: explain the model
Useful fraction 60%; outgoing branches total input.

03. Read and interpret the model
100=60+25+15; balanced.

Model / diagram check



04. Apply the physics
Initial 16 J; ideal 24 J; with thermal transfer final 19 J.

05. Explain your reasoning
Need partition, entry losses, ejecta, air blast, deformation, radiation and coupling; claim is unsupported.

06. Review: connect an earlier idea *Review: Unit 4, Day 9.*
ΔU=+20 J; gained. Less negative means greater energy.

07. Exit check
10 J other; 75% useful.

08. STRETCH: test the model
E.g. thermal 15, other 10; or thermal 20, other 5 J. Both balance but are hypotheses.

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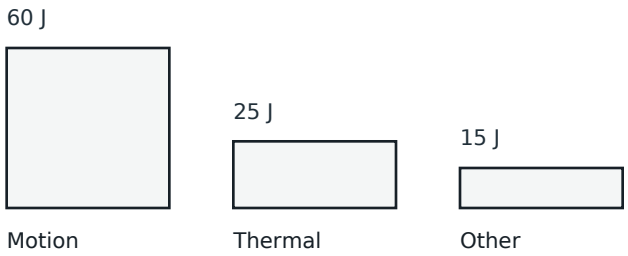
Accept equivalent correct methods and clearly labeled models. Language supports do not change the physics demand. Supplied data are examples; actual lab measurements and teacher checkpoints must be recorded from the lesson.

01. Warm-up: bring back an earlier idea *Review: Unit 4, Day 18.*
Energy=3.75 J; average opposing force≈5.357 N.

02. Worked example: explain the model
Useful fraction 60%; outgoing branches total input.

03. Read and interpret the model
100=60+25+15; balanced.

Model / diagram check



04. Apply the physics
Initial $+2\times10^{6}$ J; after -1×10^{6} J. Negative energy is bound in ideal isolated gravity model; positive permits unbound trajectory with conditions.

05. Explain your reasoning
Need partition, entry losses, ejecta, air blast, deformation, radiation and coupling; claim is unsupported.

06. Review: connect an earlier idea *Review: Unit 4, Day 9.*
U_i=-48 J; U_f=-24 J; ΔU=+24 J; gravity work=-24 J.

07. Exit check
Initial 60 J; K 48 J; $v=\sqrt{48}=6.93$ m/s.

08. STRETCH: test the model
Input 3000 J; useful 1200 J; efficiency 40%; remaining 1800 J.